

Exposure to air and water pollution during pregnancy and risk of adverse perinatal outcomes in São Paulo State: protocol for a retrospective population-based cohort study

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ABSTRACT

Introduction The fetus is highly susceptible to extrauterine environmental influences during its development. Exposure to air and water pollutants may compromise fetal growth, increasing risks of morbidity and mortality. Since such exposure can be mitigated through governmental interventions, it is necessary to deepen our understanding of pollution's potential impact in this context. This protocol proposes a study which aims to investigate the association between maternal exposure to air and water pollutants during pregnancy and adverse perinatal outcomes (preterm delivery, low birth weight, small-for-gestational-age births, congenital anomalies and stillbirth) in São Paulo State, Brazil, between 2015 and 2020.

Methods and analysis We will carry out a retrospective population-based cohort study to assess differences in rates of preterm delivery, low birth weight, small-for-gestational-age births, congenital anomalies and stillbirths. All births will be classified according to gestational exposure to air pollutants (particulate matter, nitrogen dioxide, sulfur dioxide, ozone) and water pollutants (nitrate-nitrogen, phosphorus, manganese, nickel, zinc, cadmium, mercury, lead, aluminium, iron and copper). Individual and environmental data will be retrieved from public databases. Inferential analysis will be conducted through a Mixed-Effects Model with prespecified sensitivity analysis. Furthermore, as a secondary objective, we will compare traditional statistical approaches with deep learning methods.

Ethics and dissemination The study will use publicly available, deidentified data. Formal ethical approval is not required, according to the Brazilian National Commission of Ethics in Research (CONEP). Findings will be disseminated through conferences and peer-reviewed publications.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Air and water pollution are major public health concerns, as widespread pollutants can adversely affect human health. The full impact of these pollutants on obstetric outcomes remains difficult to quantify.

WHAT THIS STUDY ADDS

⇒ Brazil has a unique geographic and economic profile that shapes pollution exposure in ways distinct from what is currently documented elsewhere. However, robust data analyses linking air and water pollution to obstetric outcomes are still lacking for this country. This study provides a protocol to analyse data in this setting.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Analysing the available data in this context could offer valuable insights into these complex relationships and create opportunities to improve obstetric outcomes in developing countries.

INTRODUCTION

Intrauterine life represents a period of heightened vulnerability to environmental exposures, with potential consequences for both short and long-term neonatal health. Environmental pollution is a major contributor to disease and premature mortality, and the fetus is particularly susceptible to toxic exposures.^{1 2}

Human exposure to pollutants occurs through multiple pathways, including environmental tobacco smoke, outdoor air pollution from multiple sources, indoor air pollution from solid fuel use and water contamination by chemicals, disinfection

by-products and metals. Many widely used substances remain insufficiently tested for safety regarding children's health.^{1 2}

Most studies on air pollution have focused on particulate matter (PM), sulfur dioxide (SO₂), nitrogen dioxide (NO₂) and ozone (O₃). PM is a heterogeneous mixture of particles classified by aerodynamic diameter, commonly as PM₁₀ (<10 µm) and PM_{2.5} (<2.5 µm). Evidence suggests associations between PM₁₀ exposure and spontaneous miscarriage, as well as between PM₁₀ and PM_{2.5} exposure and stillbirth.³ Limited evidence also indicates an association between elevated PM₁₀ and PM_{2.5} levels and low birth weight, as well as higher rates of preterm birth and PM₁₀.⁴ Findings regarding SO₂ remain inconsistent,³ while O₃ exposure has been associated with preterm birth in a Chinese cohort.⁵ Importantly, adverse outcomes related to air pollution are often compounded by socioeconomic disadvantage, reinforcing pollution's role as both a consequence and driver of poverty.^{6 7} Thus, pollution is not only associated with poverty as a consequence but also as a causal factor by inducing morbidity and premature death.² It is still unclear if there are possible relationships between certain substances and specific timeframes during pregnancy as well as possible impacts in different categories of preterm delivery (PTD) and low birth weight.⁸

Water pollution also poses risks to obstetric outcomes. Common contaminants include chemical agents (eg, arsenic, atrazine, lead, nitrate), radionuclides (eg, radon, uranium, radium) and microbial/disinfection by-products (eg, trihalomethanes, haloacetic acids and faecal indicator bacteria).⁹ Associations have been reported between nitrate contamination and increased risk of preterm birth and low birth weight.^{9 10}

Existing studies face limitations, particularly as most research has been conducted in developed countries, where pollutant levels are often lower than in low- and middle-income countries. Thus, further research is needed to clarify associations between pollutants and perinatal outcomes in diverse contexts.

Additionally, findings from this research may reveal the potential impact of public ecological interventions on regional public health, guiding future research and public policies.

This study protocol proposes to assess the possible association between mean annual exposure to air and surface water pollutants and adverse perinatal outcomes (preterm birth, low birth weight, small-for-gestational-age birth, congenital anomalies and perinatal mortality) in São Paulo State from 2015 to 2020. Furthermore, we aim to benchmark traditional inferential analyses against deep learning (DL) approaches in this complex setting of multiple exposures and outcomes.

METHODS

Study design

We will carry out a retrospective cohort study using ecological exposure assignment to investigate the potential association between air and water pollution and adverse perinatal outcomes. This study is scheduled to begin in July 2026 and is expected to span 2 years, during which its findings will be disseminated through academic conferences and peer-reviewed journals.

Setting

São Paulo State, one of Brazil's 27 federative units, comprises 645 municipalities, covering a total area of 248,219.481 km² and housing a population of 44 420 459 inhabitants, according to the Brazilian Institute of Geography and Statistics (IBGE). In 2010, its Human Development Index was 0.783, ranging from 0.639 to 0.862 (IBGE, 2023). [Figure 1](#), produced by the IBGE and adapted by the authors, shows Brazil's area and population density of Brazil (A) and those of São Paulo State (B) in 2022.

Data will be used regarding births occurring in São Paulo State from 2015 to 2020 with at least one known exposure variable, as identified by the São Paulo State Environmental Company, which has information collection points distributed in different cities in the state ([figure 2](#)).

Participants

Inclusion criteria

We will include data on all live births and stillbirths in São Paulo State between 2015 and 2020 for individuals who live in municipalities with available measurement of air and surface water pollutants. Births will be considered eligible if gestational age was ≥20, or if birth weight was

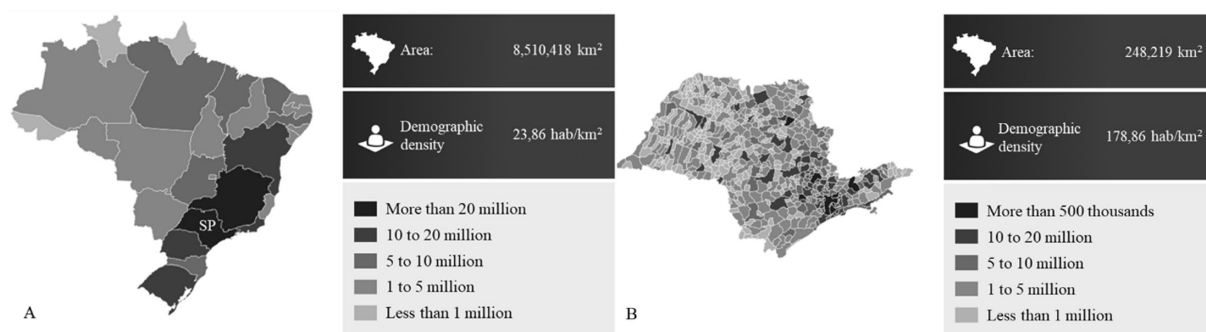


Figure 1 The Brazil (A) and São Paulo State (B) area and demographic density. (IBGE, 2023). IBGE, Brazilian Institute of Geography and Statistics.

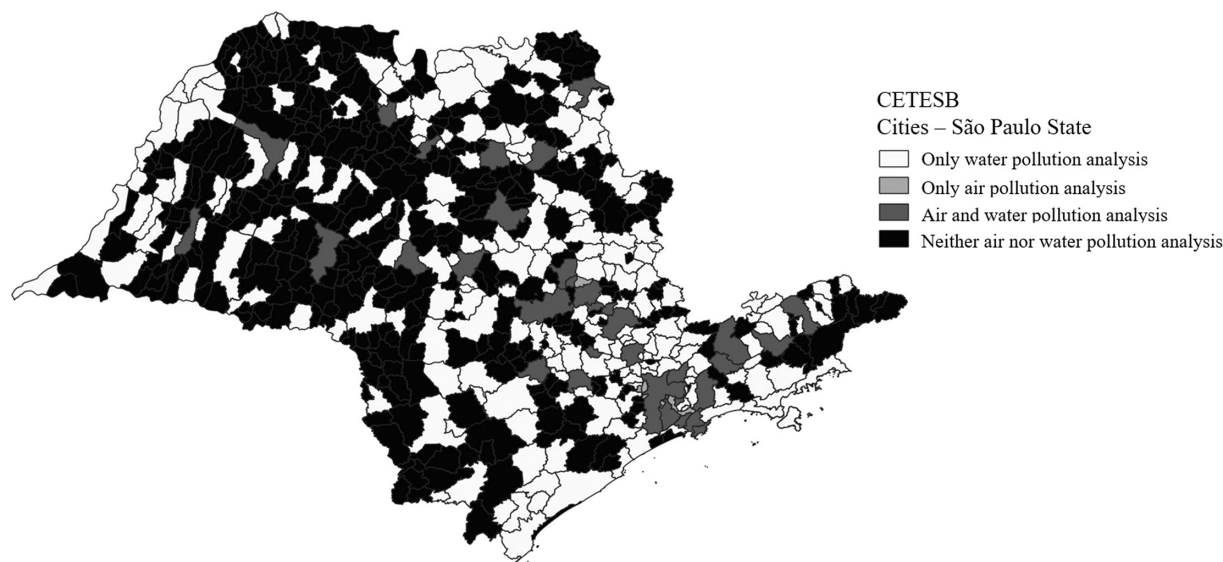


Figure 2 Distribution of information collection points on air and water pollution in the state of São Paulo, according to CETESB between 2015 and 2020. CETESB, São Paulo State Environmental Company.

≥500 g. For each municipality and year, average annual levels of air pollutants (PM <10µm and <2.5µm, NO₂, SO₂, O₃) and water pollutants (nitrate-nitrogen, phosphorus, manganese, nickel, zinc, cadmium, mercury, lead, aluminium, iron and copper) will be included.

Exclusion criteria

We will exclude individuals (1) if no valid measurements of the selected exposure variables exist in the municipality of residence during pregnancy. Measurements will only be considered valid if deemed representative by CETESB and not obtained by proxy (ie, from a monitoring station located in another municipality) and (2) if the city of residence or pregnancy length is unknown.

Summary of participants

In 2020, São Paulo State had over 46million residents. Of its 645 municipalities, 394 had no measurement for either air or water pollution, representing 15% of the state's population. Six municipalities have data exclusively concerning air pollution, while 210 had data solely regarding water pollution. 35 municipalities—home to 54% of the State's population—had data on both air and water pollution. Based on these criteria, 2 349 161 deliveries between 2015 and 2020 are eligible for analysis under this protocol.

Variables

Exposure variables

Mean annual levels of air and surface water pollutants are presented in [table 1](#). Brazil's National Air Quality Standards were last revised under Resolution 491/2018.¹¹ This regulation is legally enforceable and is defined by the National Environmental Council (CONAMA), an official government agency which sets enforceable limits for pollutants. These standards include definitive values (annual arithmetic means) and intermediate values

based on the WHO guidelines from 2005. Although WHO has since updated its recommendations,¹² this study will focus on exposures exceeding Brazilian legal thresholds to assess their impact in this setting. For all variables, the definitive air quality value was used, except for SO₂, which only has intermediate values. O₃ was reported as the percentage of measurements exceeding the intermediate threshold, as no definitive annual mean was available. CONAMA also sets standards for water quality and acceptable water pollutants, under Resolution 357/2005,¹³ amended by Resolution 430/2011. Unlike air pollutants, water pollutant limits vary by waterbody type, as defined by State Decree 10 755/1977. These water sources are considered suitable for human consumption, though treatment requirements differ by classification. Annual arithmetic means were available for nitrate-nitrogen and total phosphorus, while other water pollutants were reported as the percentage of non-conforming measurements.

Assignment of exposure

Two variables will be used to define pollutant exposure levels. First, we will rely on the city of residence reported on birth and fetal death certificates. For congenital anomaly outcomes, exposure will be assigned based on the first trimester of pregnancy; for all other outcomes, the whole pregnancy period will be considered. Since these exposure windows may overlap two calendar years, data from both years will be included and weighted according to the amount of time spent in each year. The city of residence and year of exposure will then be matched to the corresponding levels of air and surface water pollutants. Whenever the same town, within the same year, has multiple stations reporting different values for the same contaminant, all measurements will be incorporated using a clustering-based statistical methodology.

Table 1 Mean annual levels of air and surface water pollutants according to acceptable and non-acceptable levels

	Acceptable up to	Non-acceptable
Air pollutants		
Fine particulate matter (PM _{2.5})	10 µg/m ³	>10 µg/m ³
PM ₁₀	20 µg/m ³	>20 µg/m ³
Sulfur dioxide	20 µg/m ³	>20 µg/m ³
Nitrogen dioxide	40 µg/m ³	>40 µg/m ³
Ozone (O ₃)*		
São Paulo State standard	140 µg/m ³	>140 µg/m ³
National standard	160 µg/m ³	>160 µg/m ³
Water pollutants		
Nitrate-nitrogen	10 mg/L	>1.0 mg/L
Total phosphorus	0.15 mg/L	>0.15 mg/L
Total manganese**	0.1–0.5 mg/L	>0.5 mg/L
Total nickel**	0.025 mg/L	>0.025 mg/L
Total zinc**	0.18–5 mg/L	>0.18–5 mg/L
Total cadmium**	0.001–0.01 mg/L	>0.01 mg/L
Total mercury**	0.0002–0.002 mg/L	>0.002 mg/L
Total lead**	0.01–0.033 mg/L	>0.033 mg/L
Soluble aluminium**	0.1–0.2 mg/L	>0.2 mg/L
Soluble iron**	0.3–5 mg/L	>5 mg/L
Soluble copper**	0.009–0.013 mg/L	>0.013 mg/L

The final value is expressed as the percentage of measurements that exceed these limits.

*O₃ is reported as the percentage of non-compliant measurements. These measurements are evaluated against two standards: the São Paulo state standard, which requires an average of 8 consecutive hours below 140 µg/m³, and the national standard, which requires a 1-hour average below 160 µg/m³. **Expressed as a percentage of non-conforming measures.

Summary of known exposure values

Applying the criteria described above, we calculated expected exposure levels for the proposed cohort. As previously stated, 251 municipalities—representing approximately 85% of the State's population—have at least some pollution measurement available. For municipalities with multiple monitoring stations in a given year, mean values were used to determine exposure levels. A summary of pollutants with continuous data availability, along with their mean values and SDs, is presented in [table 2](#).

Confounding variables

The number of potential confounding variables available for analysis is limited, as public databases provided by the Brazilian government do not include more detailed information. Therefore, we will use available variables such as (1) newborn sex; (2) maternal age group at birth, categorised as <20 years, 20–35 years and >35 year; (3) maternal education level, classified as ≤8 years, 9–11 years and ≥12 years of education and (4) decomposed gross domestic product (GDP) per capita, divided into agricultural, industrial, non-public services and public services. Exposure to GDP will be defined in the same way as pollutant exposure, using the city of residence and pregnancy duration to determine location and time.

When pregnancies span two calendar years, GDP values will be weighted according to the proportion of time in each year. Nominal GDP values for each component will be used for 2019, the last year of the study, while earlier values will be adjusted using Brazil's Extended National Consumer Price Index (IPCA). IPCA adjustments are calculated based on values from the first month of every year and are publicly available at the IBGE. Nominal GDP per capita values for each component are also provided by the IBGE and are expressed in Brazilian currency (Real).

Outcome measures

We defined the primary outcomes as (1) small for gestational age (SGA), as defined by Alexander *et al*¹⁴; (2) PTD; (3) stillbirth and (4) major congenital anomaly. Live birth certificates include a field indicating congenital abnormality, recorded as present, absent or ignored, as well as a field for the International Classification of Diseases, 10th Revision (ICD-10) code. For stillbirths, ICD-10 codes are recorded for cause of death and contributing factors. ICD-10 codes contain up to three digits following the chapter letter. These codes will be used to classify each case to major congenital anomaly groups and system subgroups according to the EUROCAT 1.5 Guide.¹⁵

Table 2 The mean annual levels of air and surface water pollutants recorded in municipalities of São Paulo State from 2015 to 2020

	Mean (SD)
Air pollutants	
Fine particulate matter (PM _{2.5})	16.8 (1.5) µg/m ³
PM ₁₀	28.5 (4.35) µg/m ³
Sulfur dioxide	2.73 (1.67) µg/m ³
Nitrogen dioxide	31.4 (8.67) µg/m ³
Ozone*	0.485 (0.90)%
Water pollutants	
Nitrate-nitrogen	0.789 (0.880) mg/L
Total phosphorus	0.89 (0.696) mg/L
Total manganese*	23.7 (19.7)%
Total nickel*	1.85 (6.08)%
Total zinc*	0.481 (2.72)%
Total cadmium*	0.134 (1.16)%
Total mercury*	0.575 (2.87)%
Total lead*	0.877 (4.03)%
Soluble aluminium*	23.6 (22.9)%
Soluble iron*	42.3 (29)%
Soluble copper*	7.48 (8.72)%

*Values expressed as the mean percentage of non-compliant measurements, since continuous monitoring data for these pollutants are not available.

Level of variable measurement

As noted, this study is a retrospective cohort with ecological exposure attribution. Table 3 summarises all study variables according to type and level of measurement.

Data sources

Data will be obtained from publicly available sources. Information on live births will be retrieved from the Live Birth Information System (SINASC), while data on stillbirths will be extracted from the Mortality Information System (SIM), both made publicly available by Brazil's Ministry of Health. Data on air and surface water pollutant levels in São Paulo State will be collected from annual reports published by the State of São Paulo Environmental Company (CETESB). Municipal-level GDP components will be obtained from the IBGE. Details regarding all databases used in this study are summarised in table 4.

The role of SIM and SINASC in documenting congenital anomalies requires further clarification. These systems rely on birth certificates—whether for live birth or stillbirth—which are consolidated immediately or shortly after the delivery. Thus, they do not capture information that may only become available months later, as is the case in some countries. Their records are limited to data obtained during antenatal care and immediately after birth. Regarding live births, physicians record any observed anomalies in a descriptive field, which is subsequently coded into ICD-10 classifications by specifically trained professionals.

DATA ANALYSIS

Proposed analysis

The primary objective of this study is to evaluate if different air and water pollutants influence major obstetric outcomes, including PTD, small-for-gestational-age birth, stillbirth and major congenital anomalies. A secondary objective is to compare the results obtained through traditional inferential methods with those derived from DL approaches.

Table 3 Description of variables in the protocol

Variable	Type of variable	Level of measurement
Air pollution measurements	Assessed as binary (above threshold or not) in all cases; and as continuous, in case of pollutants with available continuous data.	Municipal level
Water pollution measurements	Assessed as binary (above threshold or not) in all cases; and as continuous, in case of pollutants with available continuous data.	Municipal level
Newborn sex	Binary (male or female)	Individual level
Maternal age group	Categorical, according to years of education	Individual level
Gross domestic product components	Continuous	Municipal level
Birth weight group	Binary (small for gestational age or not)	Individual level
Gestational age at delivery group	Assessed twice: as binary (preterm or term delivery) and as ranked (very preterm, preterm and term delivery)	Individual level
Birth vitality condition	Binary (stillbirth and live birth)	Individual level
Major congenital anomaly (CA) status	Binary variable (present or absent)	Individual level
Major CA per anatomic site	For each CA subgroup, a binary variable will describe it as present or absent, according to EUROCAT criteria. A single individual could have more than one subgroup presenting birth defects.	Individual level

Table 4 Description of data sources

Name of database	Content of database	Location of database
SINASC	Individual-level demographic and clinical information for all live births	https://datasus.saude.gov.br/transferencia-de-arquivos/
SIM	Individual-level demographic and clinical information for all deceased individuals	https://datasus.saude.gov.br/transferencia-de-arquivos/
CETESB	Aggregated annual information for pollutant levels	https://cetesb.sp.gov.br/
IBGE	GDP components per capita in each city in the state of São Paulo, in nominal values, per year	https://www.ibge.gov.br/estatisticas/economicas/contas-nacionais/9088-produto-interno-bruto-dos-municipios.html?=&t=resultados
IBGE Inflation Calculator	Online calculator for monetary inflation in Brazil	https://www.ibge.gov.br/explica/inflacao.php

CETESB, State of São Paulo Environmental Company; GDP, gross domestic product; IBGE, Brazilian Institute of Geography and Statistics; SIM, Mortality Information System; SINASC, Live Birth Information System.

Pollutant data will serve as the exposure variable, assessed for associations with the following outcomes: small-for-gestational-age births, PTD, stillbirth and birth defects. PTD will be analysed both as a binary and ranked variable, distinguishing very PTD (<32 weeks) from PTD (PTD <37 weeks). Stillbirth will also be examined as both a binary variable and a ranked variable, differentiating early stillbirth (<28 weeks) from late stillbirth (≥28 weeks).

Exploratory interaction effects will also be investigated. Previous studies have reported a higher prevalence of small-for-gestational-age newborns among those with congenital anomalies^{16 17}; this interaction will also be tested. Associations have also been identified between congenital anomalies and preterm birth,¹⁷ as well as between stillbirth and small-for-gestational-age newborns¹⁸; these interactions will likewise be explored. Finally, GDP per capita will be included in all models, given its documented correlation with pollutant levels and its potential mitigating effect on pollution-related adverse outcomes.^{19 20}

Bias

Several potential confounders may affect the study outcomes, including maternal age, maternal body mass index, obstetric comorbidities, antenatal care, partnership status and education level.^{21–26} Not all of these variables are directly identifiable in public databases. Moreover, as this is a retrospective study using administrative data, inaccuracies in record-keeping are possible. The exposure variable is designed to be ecologically attributed, which may introduce random misclassification. Additionally, pollution may be associated with economic activity, which, in the short run, could generate economic gains and improve perinatal outcomes. Taken together, these factors suggest a bias towards underestimating the true results.

Study size

Based on inclusion and exclusion criteria, the study population comprises 2 349 161 individuals. Preliminary analysis identified 272 170 (11.6%) preterm births; 314 762 (13.4%) small-for-gestational-age births; 17 946 (0.8%) stillbirths and 33 761 (1.4%) cases of congenital malformations (not exclusively major). For comparison, national rates are 11.1% for PTD²⁷; 9.2% for small-for-gestational-age births²⁸; 8.3 cases of congenital anomalies per 1000 live births²⁹ and 8.34 stillbirth cases per 1000 deliveries in the southeastern region,³⁰ where São Paulo State is located. This sample size is sufficient, considering that fewer than ten variables will be included in the logistic regression models to minimise overfitting.

Statistical analysis

The analysis will include both descriptive and inferential methods to assess the relationship between pollutant exposure and adverse perinatal outcomes. Variables will be described according to their type: categorical variables by frequencies and percentages, and continuous variables by means and SD, if normally distributed, or medians and IQRs otherwise. Inferential analysis will use a mixed effects model (MEM) to adjust for air and water pollutants while controlling for potential confounders, which constitutes the primary aim of this research. Participants will be clustered at the municipality level to account for multiple pollution measures per city. Two sensitivity analyses are planned: (1) a fixed effects model (FEM) using the same dataset and (2) a MEM incorporating multiple imputation methods to address missing data. A summary of the proposed inferential analysis is presented in [table 5](#).

As previously noted, a secondary objective of this study is to compare traditional inferential methods with DL approaches. To this end, we will use deep neural networks (DNN) to identify complex and non-linear patterns between pollutant exposure and perinatal outcomes.

Table 5 Planned statistical analysis

Objective			
To investigate the association between:	Outcome	Exposure	Method of analysis
Air pollutants and SGA	SGA	Air pollutants	MEM, FEM, DNN, SEM, PSM*
Air pollutants and PTD	PTD	Air pollutants	MEM, FEM, DNN, SEM, PSM†
Air pollutants and stillbirth	Stillbirth	Air pollutants	MEM, FEM, DNN, SEM, PSM‡
Air pollutants and CA	CA	Air pollutants	MEM, FEM, DNN, SEM, PSM§
Water pollutants and SGA	SGA	Water pollutants	MEM, FEM, DNN, SEM, PSM*
Water pollutants and PTD	PTD	Water pollutants	MEM, FEM, DNN, SEM, PSM†
Water pollutants and stillbirth	Stillbirth	Water pollutants	MEM, FEM, DNN, SEM, PSM‡
Water pollutants and CA	CA	Water pollutants	MEM, FEM, DNN, SEM, PSM§

*Adjusted for newborn sex, maternal age group, education level, GDP per capita, stillbirth and congenital anomaly.

†Adjusted for newborn sex, maternal age group, education level, GDP per capita and congenital anomaly.

‡Adjusted for newborn sex, maternal age group, education level, GDP per capita, congenital anomaly and SGA.

§Adjusted for newborn sex, maternal age group, education level, GDP per capita and SGA.

CA, congenital anomaly; DNN, deep neural network; FEM, fixed effects model; GDP, gross domestic product; MEM, mixed effects model; PSM, propensity score matching; PTD, preterm delivery; SEM, structural equations model; SGA, small for gestational age.

DNN will be trained on a training dataset and validated using a testing dataset to predict the study's proposed outcomes. We will also use autoencoders to detect anomalies in the exposure data, potentially revealing areas or periods of time of extreme pollution that require further investigation. Structural equations model (SEM) will allow us to evaluate both direct and indirect causal pathways between exposure and outcomes, accounting for multiple mediating and moderating variables. To reduce selection bias, propensity score matching (PSM) will be implemented to match exposed and non-exposed individuals based on relevant covariates.³¹ Finally, confounding sensitivity analysis (E-values) will be conducted to evaluate the robustness of our findings against potential unmeasured confounders.³²

Model construction

Exposure variables (air and water pollutants) will be modelled both as continuous and binary variables, depending on data availability and analytical purpose. Binary exposure will be defined according to the thresholds presented in table 1. For pollutants expressed as percentages of non-conforming measures, exposure will be defined as the presence of any non-conforming measurement.

Separate models will be constructed for each outcome, distinguishing between air and water pollutants. All models will be adjusted for predefined confounders, including newborn sex, maternal age group, maternal education level and decomposed GDP per capita, as described in table 4.

MEM will serve as the primary analytical approach, incorporating municipality-level random effects to account for clustering. FEM will be used as sensitivity analyses.

For machine learning (ML) approaches, input features will include exposure variables and confounders. Feature

preprocessing will involve normalisation of continuous variables and encoding of categorical variables. Models will be designed to enable direct comparison between traditional statistical approaches and ML methods under consistent input configurations.

Interaction terms between exposure variables and selected covariates will be explored in both statistical and ML models to assess potential effect modification. This modelling framework ensures consistency across analytical strategies while enabling comparison between linear, non-linear and causal modelling approaches.

Causal analysis with ML models

ML, particularly DL, provides powerful tools to identify complex patterns in large datasets, making it especially suitable for analysing multifaceted relationships between environmental exposures and perinatal outcomes. DL models, such as DNN, can automatically learn hierarchical data representations,³³ thereby detecting subtle interactions and non-linear effects among observed variables.

To incorporate DL into our causal analysis, we will use a structural causal model (SCM).³⁴ SCM provides a formal and interpretable framework for representing causal relationships, enabling explicit modelling of mechanisms underlying observed associations. A directed acyclic graph, informed by domain knowledge, will represent relationships among environmental pollutants, maternal factors and perinatal outcomes, facilitating causal effect estimation and interpretation.^{35 36}

The dataset will be divided into training (70%), validation (15%) and testing (15%) subsets. To prevent data leakage, observations from the same municipality will remain within the same subset. Model performance will be evaluated on the held-out test set. Additionally, k-fold cross-validation (k=5) will be conducted within the training data to assess model stability and mitigate

overfitting, while hyperparameters will be optimised using the validation set.

Input features will include pollutant exposure variables, maternal sociodemographic characteristics and other covariates described in this study. Continuous variables will be normalised, categorical variables will be encoded using appropriate techniques and missing data will be addressed using multiple imputation strategies consistent with the statistical analysis.

DNN architectures will be tailored to dataset characteristics and trained to estimate relationships between exposures and outcomes. These models will support causal inference through approaches such as counterfactual prediction and representation learning.³⁷ Advantages of DL in this context include the ability to capture non-linear relationships, handle high-dimensional data, mitigate confounding effects and extract insights about feature importance, thereby contributing to the identification of potential causal drivers.³⁸

To enhance interpretability, attention mechanisms and feature attribution methods will be incorporated to highlight the most relevant exposure variables and critical periods during pregnancy.

In addition to DL-based approaches, complementary methodologies such as the Aleph approach³⁹ will be applied to identify temporal causal relationships and lag structures between exposure variables and outcomes. This approach emphasises causal drivers through feature importance and temporal dependencies, while accounting for confounding variables.

By integrating DL models with SCM and complementary causal discovery methods, this framework enables robust effect estimation while capturing complex interactions and temporal dynamics inherent in the data.

Patient and public involvement

This research will not involve any patients directly, as this protocol is based on retrospective analysis. Public engagement will occur through dissemination in academic conferences, scientific reporting and institutional media initiatives.

DISCUSSION

To the best of our knowledge, this study represents the most extensive population-based investigation proposed in Brazil on the association between adverse perinatal outcomes and maternal exposure to air and water pollutants during pregnancy.

We acknowledge that the limitation lies in information and confounding bias inherent to retrospective population-based cohorts using a public database. Although efforts are made to ensure accuracy, public databases may contain errors or limitations in the data collected. Furthermore, the governmental database provides only a limited set of potential confounding variables. Nevertheless, given the large population size, factors such as smoking and physical activities

may be similarly distributed across groups. Since unknown confounders are unlikely to be systematically distributed, but rather randomly, we expect these limitations to attenuate rather than exaggerate any observed associations.

These factors also limit the conclusions that can be drawn from DL strategies. While this methodology can help develop causal reasoning about how these exposures might affect the outcomes under study, it should not be interpreted as establishing definitive causal relationships. It can, however, contribute to a better understanding of how complex variables correlate and help guide future research on the topic.

Furthermore, regarding air pollution, our research focuses exclusively on outdoor air pollution. Both indoor and outdoor air pollution are recognised as major contributors to health outcomes. It is estimated that individuals in developed countries spend 90% of their time indoors.⁴⁰ Studies have shown that indoor air quality can differ substantially from outdoor air quality, depending on human behaviour, ventilation, heating methods and other factors.^{41 42} This limitation has also been noted in other registry-based studies assessing the impact of air pollution on pregnancy-related outcomes.^{43 44}

Despite the technological, infrastructural and legal challenges of using large-scale secondary data, Brazil offers a robust regulatory environment, high-quality medical records, an expanding network of data linkage centres and a transparent, up-to-date approach for research and policy including health technology assessment.^{45 46}

Adverse perinatal outcomes such as PTD, low birth weight, congenital anomalies and stillbirth impose significant burdens on individuals, society and public health. Public health initiatives and regulations aimed at reducing these outcomes can be more effective when informed by evidence on the potential impact of air and water pollution.

If an association between pollution and adverse perinatal outcomes is identified in São Paulo State and the pollution levels are modifiable, government interventions could be implemented to reduce exposure and improve maternal and fetal health. Such evidence can guide public health actions and strategies.

This research also highlights issues of environmental injustice, particularly in socioeconomically disadvantaged and industrial areas, where vulnerable populations may be disproportionately exposed. Findings can inform more equitable resource distribution and environmental protection policies. Additionally, this study introduces the use of ML techniques to uncover complex, non-linear relationships between pollutant exposure and adverse perinatal outcomes. These methods complement traditional analyses by enhancing predictive accuracy and identifying anomalies in pollution data. While promising, it remains

essential to address algorithmic biases and ensure model transparency to promote equitable health outcomes across populations.

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Contributors CGFF and VCM were involved with the research question and project conceptualisation. NJWMJ, LBBGM, JMME, CGFF, JFA, LT and VCM designed the study methods. LT reviewed the methods. SB reviewed the statistics analysis. NJWMJ, LBBGM, CGFF, JMME and VCM wrote the original draft. JFA, LT, DCCS, LADLF and VCM reviewed and edited the paper. JFA, LT and VCM supervised the research group. VCM acts as guarantor for the study.

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Ethics approval All results will be disseminated through conferences and peer-reviewed publications. The final manuscript will comply with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines. The study will rely on a database of publicly available and anonymous information. According to the Brazilian National Commission of Ethics in Research (CONEP), research using public domain data that does not identify participants, or literature reviews without human involvement does not require ethics board approval.⁴⁷ Regarding the use of Machine LearningML techniques, we are committed to ensuring fairness in our model by addressing potential algorithmic biases and promoting transparency. To achieve this, we will implement methods that enhance model explainability, enabling the identification and mitigation of disparities in outcomes and fostering equitable results across diverse populations.

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